

[PHYS 6268] Nonlinear Dynamics: Problem Set 7

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Strogatz Book: Problems 7.1.6, 7.2.9, & 7.3.1

1 Problem 7.1.6: Circuit for van der Pol Oscillator

Figure 1 shows the tetrode multivibrator circuit used in the earliest commercial radios and analyzed by van der Pol. In van der Pol's day, the active element was a vacuum tube; today it would be a semiconductor device. It acts like an ordinary resistor when I is high, but like a negative resistor (energy source) when I is low. Its current-voltage characteristic $V = f(I)$ resembles a cubic function, as discussed below.

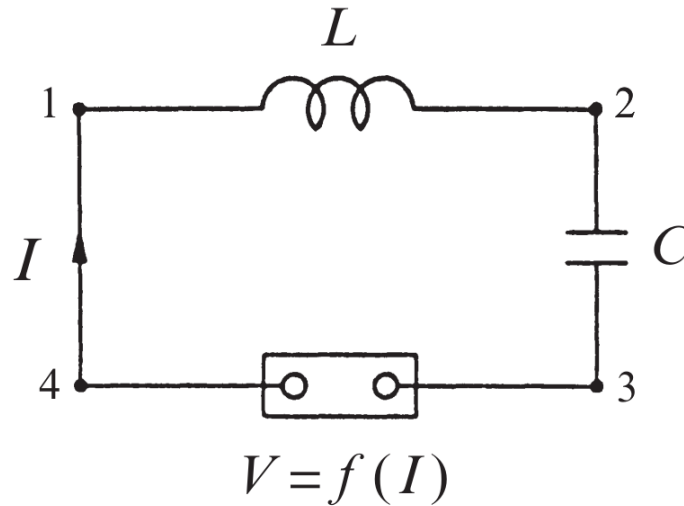


Figure 1: Circuit diagram for problem 7.1.6. Adopted from Figure 7.1.6.1 in Strogatz.

Suppose a source of current is attached to the circuit and then withdrawn. What equations govern the subsequent evolution of the current and the various voltages? Let $v = V_{32} = -V_{23}$ denote the voltage drop from point 3 to point 2 in the circuit.

1.1 (a) Show that $\dot{V} = -I/C$ and $V = LI + f(I)$.

The circuit under investigation here (see Figure 1) is a closed loop, or a series circuit. Hence, it is subject to Kirchhoff's voltage law, which states that the sum of voltage drops in a series circuit is equal to the total of all voltage sources in the loop. This follows directly from Faraday's law, $\nabla \times \mathbf{E} + \dot{\mathbf{B}} = 0$. Adopting the notation above, the voltage drops between each of the four corners are labeled $V_{12} = -V_{21}$, $V_{34} = -V_{43}$ and $V_{41} = -V_{14}$. Then, from Kirchhoff's voltage law, it is given that

$$\sum_{i=1}^3 V_{i,i+1} + V_{41} = 0 \quad .$$

The first segment we consider is that of the capacitor, corresponding to the voltage $V_{23} \equiv -V$. The capacitance (C) directly relates the the voltage to the charge Q of the capacitor:

$$Q = CV_{23} \quad .$$

Then, using $I = \dot{Q}$, we can write

$$I = \frac{d}{dt}(CV_{23}) = C\dot{V}_{23} \quad ,$$

using the notion that the capacitance is constant. Now, replacing $V_{23} \mapsto -V$, we arrive at the expression

$$\dot{V} = -\frac{I}{C} \quad . \quad (1)$$

There is no voltage drop attributed to the circuit segment where the current source is attached, so assuming the wire itself to be resistance-free, we have that $V_{41} = 0$. Additionally, letting $V_{23} \mapsto -V$ yields a simplified form for the Kirchhoff voltage law:

$$V_{12} - V + V_{34} = 0 \quad \implies \quad V = V_{12} + V_{34} \quad .$$

Then, we can determine an expression for V by obtaining V_{12} and V_{34} . The voltage drop across an inductor (inductance L) is proportional to the rate of change of the applied current, given by

$$V_{12} = L\dot{I} \quad .$$

Then, by definition, $V_{34} = f(I)$. Hence, these voltage drops amount to

$$V = L\dot{I} + f(I) \quad . \quad (2)$$

Hence, equations (1) and (2) combine to form the system

$$\begin{aligned} \dot{V} &= -\frac{I}{C} \\ \dot{I} &= \frac{V - f(I)}{L} , \end{aligned}$$

consistent with the prompt.

1.2 (b) Show that the equations in (a) are equivalent to the van der Pol system, in a special case.

To do so, we apply the following mapping to our variables in equations (1) and (2):

$$\begin{aligned} I &\mapsto x/\sqrt{L} \\ V &\mapsto w/\sqrt{C} \\ t &\mapsto \tau\sqrt{LC} . \end{aligned}$$

We start with equation (1). Substitution of the above yields

$$\begin{aligned} \frac{dV}{dt} &= \frac{dw/\sqrt{C}}{d\tau\sqrt{LC}} = \frac{1}{C\sqrt{L}} \frac{dw}{d\tau} \quad , \quad -\frac{I}{C} = -\frac{x/\sqrt{L}}{C} = -\frac{x}{C\sqrt{L}} \\ \implies \frac{dw}{d\tau} &= -x \quad , \end{aligned}$$

and similarly

$$\frac{dI}{dt} = \frac{dx/\sqrt{L}}{d\tau\sqrt{LC}} = \frac{1}{L\sqrt{C}} \frac{dx}{d\tau} \quad , \quad \frac{V - f(I)}{L} = \frac{w/\sqrt{C} - F(x)}{L} = \frac{w}{L\sqrt{C}} - \frac{1}{L}F(x)$$

$$\implies \frac{dx}{d\tau} = w - \sqrt{C}F(x) \quad .$$

By letting $\mu \equiv \sqrt{C}$, we have shown that the system given by equations (1) and (2) is equivalent to

$$\begin{aligned} \dot{w} &= -x \\ \dot{x} &= w - \mu F(x) \end{aligned}$$

if we let the Newtonian “dot” notation represent differentiation with respect to the scaled temporal coordinate τ (exclusively in the above expression). The function $F(x) \equiv f\left(x/\sqrt{L}\right)$ is defined in the prompt.

2 Problem 7.2.9: Gradient Systems

For each of the following systems, decide whether it is a gradient system. If so, find V and sketch the phase portrait. On a separate graph, sketch the equipotentials ($V = \text{const.}$). If the system is not a gradient system, go on to the next question.

In general, a gradient system can be written in the form

$$\dot{\mathbf{x}} = -\nabla V \quad ,$$

for some continuously differentiable scalar function V . Let

$$\dot{\mathbf{x}} = [f(x, y), g(x, y)] \quad .$$

These vector equations imply (for a 2-D system) the following relationships:

$$\begin{aligned} f(x, y) &= -\frac{\partial V}{\partial x} \\ g(x, y) &= -\frac{\partial V}{\partial y}, \end{aligned}$$

which we seek to evaluate for the following systems.

2.1 (a) $\dot{x} = y + x^2y, \dot{y} = 2xy - x$

This system is *not* a gradient system. The above equation for the x component, applied to this system, yields

$$\begin{aligned} f(x, y) &= y + x^2y = -\frac{\partial V}{\partial x} \\ \implies \partial V &= -y\partial x - x^2y\partial x \\ \implies V &= -\int y \, dx + h(y) - \int x^2y \, dx + j(y) \\ \text{arbitrarily, let } h(y) + j(y) &\longrightarrow h(y) \\ \implies V(x, y) &= -xy - \frac{x^3y}{3} + h(y). \end{aligned}$$

However, considering the second component, we find that

$$V = \int x \, dy - 2 \int xy \, dy = xy + k(x) - xy^2 \quad , \quad (3)$$

which is inherently contradictory to the x component result. Hence, this is not a gradient system.

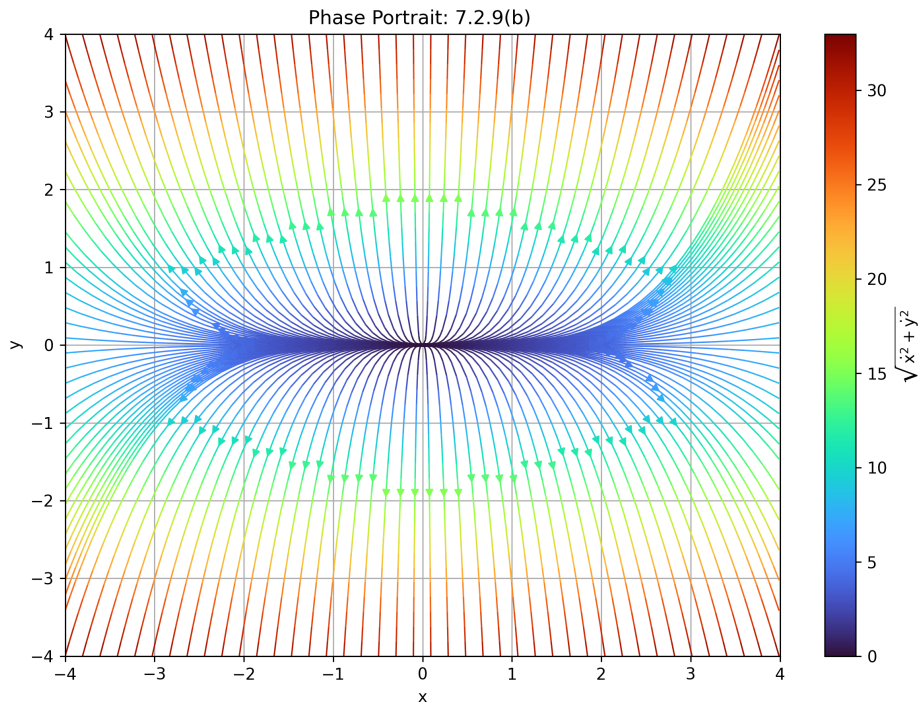


Figure 2: Phase portrait for the system $\dot{x} = 2x$, $\dot{y} = 8y$. There is a single fixed point at the origin, and it is a repeller.

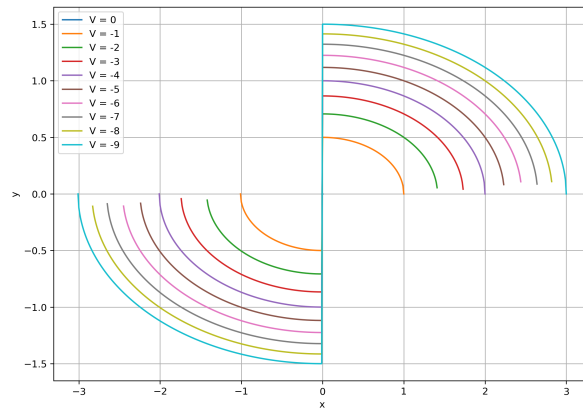


Figure 3: Isopotential curves for the gradient system $\dot{x} = 2x$, $\dot{y} = 8y$. As expected from the orientation of the trajectories in the phase portrait (Figure 2), the manifolds of constant potential V are ellipses in the phase plane.

2.2 (b) $\dot{x} = 2x$, $\dot{y} = 8y$

By inspection, $V(x, y) = -x^2 - 4y^2$. Formally, the first component yields $V(x, y) = h(y) - x^2$ for some arbitrary differentiable function h . The second component then combines with this result to produce $V(x, y) = -x^2 - 4y^2$. Hence, this is a gradient system.

The phase portrait for the system is shown in Figure 2. As indicated by the phase portrait, the lone fixed point at the origin is an unstable source, or a repelling node. Contours of the gradient system, equal to isolines of the potential $V(x, y)$, are depicted in Figure 3. The isolines are linearly spaced ellipses, consistent with the coefficients of f and g .

2.3 (c) $\dot{x} = -2x e^{(x^2+y^2)}, \dot{y} = -2y e^{(x^2+y^2)}$

Again, by inspection, the potential for this system is given by $V(x, y) = e^{(x^2+y^2)}$. The phase portrait, depicted in Figure 4 shows exponentially attracted trajectories flowing evenly towards the origin. As expected, the isolines for the potential (satisfying $V = \text{const.}$) are circles that are *logarithmically spaced*, that is, the distance between successive constant values of V decreases exponentially.

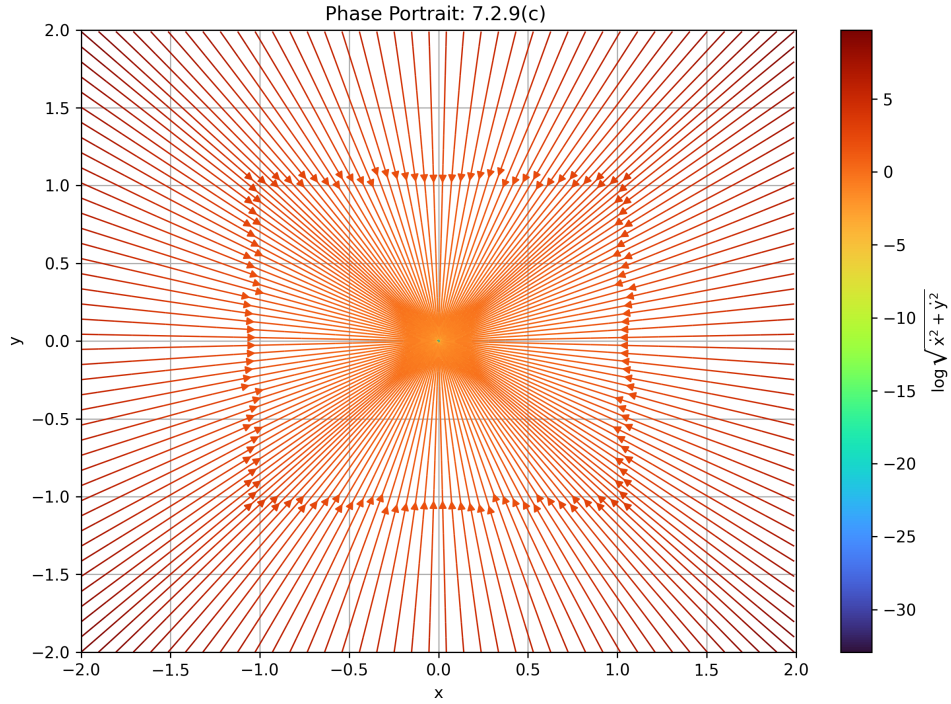


Figure 4: Phase portrait for the system $\dot{x} = -2x e^{(x^2+y^2)}, \dot{y} = -2y e^{(x^2+y^2)}$. There is a single fixed point at the origin, and it is an attractor.

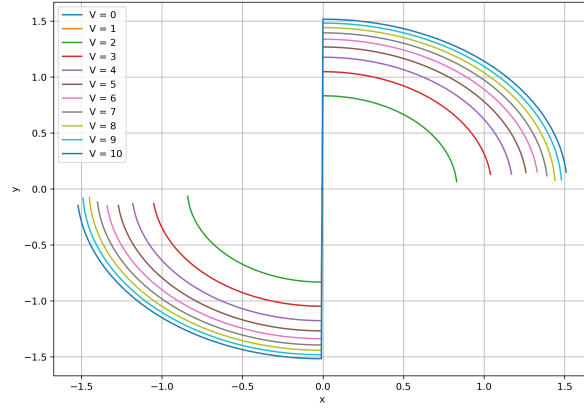


Figure 5: Isopotential curves for the gradient system $\dot{x} = -2x e^{(x^2+y^2)}$, $\dot{y} = -2y e^{(x^2+y^2)}$. As expected from the orientation and symmetry of the trajectories in the phase portrait (Figure 4), the manifolds of constant potential V are circles in the phase plane.

3 Problem 7.3.1: Poincaré–Bendixson Theorem

Consider

$$\begin{aligned}\dot{x} &= x - y - x(x^2 + 5y^2), \\ \dot{y} &= x + y - y(x^2 + y^2)\end{aligned}$$

3.1 (a) Classify the Fixed Point at the Origin

To classify the fixed point at the origin, we consider the linearized matrix of stability for the system. After calculating this matrix as a function of system variables (x, y) , we can evaluate it at the fixed point $(x^*, y^*) \rightarrow (0, 0)$ and determine the system behavior. We begin by calculating the matrix of stability.

Let $\mathbf{x}$ be a column vector containing x, y . Then, linearization of the system is given by

$$\frac{d}{dt} \begin{pmatrix} x \\ y \end{pmatrix} = \dot{\mathbf{x}} = \mathbb{A} \cdot \mathbf{x} + \mathcal{O}(\mathbf{x}^2) \quad ; \quad \mathbb{A} \equiv \begin{pmatrix} \partial_x \dot{x} & \partial_y \dot{x} \\ \partial_x \dot{y} & \partial_y \dot{y} \end{pmatrix}.$$

Evaluating the partial derivatives of the expressions for $\dot{x}, \dot{y}$ above yields

$$\mathbb{A}(x, y) = \begin{pmatrix} 3x^2 + 3y^2 + 1 & 10xy - 1 \\ 1 - 2xy & 1 - x^2 - 3y^2 \end{pmatrix}.$$

Then, it follows that

$$\mathbb{A}(0, 0) = \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}.$$

This matrix has a trace $\text{Tr } \mathbb{A} = 2$ and determinant $\det \mathbb{A} = 1 - (-1) = 2$. Since the determinant is positive, this implies that the fixed point must be a spiral or a center. To determine which, we find that the discriminant $(\text{Tr } \mathbb{A})^2 - 4\det \mathbb{A} = -4 < 0$, implying that the system has spiraling trajectories near this fixed point at the origin. Then, the eigenvalues of the matrix can be calculated from the trace and determinant:

$$\lambda_{\pm} = \frac{\text{Tr } \mathbb{A} \pm \sqrt{(\text{Tr } \mathbb{A})^2 - 4\det \mathbb{A}}}{2} = 1 \pm i, \quad (4)$$

Since $\text{Re}(\lambda_{\pm}) > 0$, these spiraling trajectories near the origin are *unstable*. Hence, the fixed point is an unstable spiral with trajectories spiraling “outward” counterclockwise.

3.2 (b) Rewrite the System in Polar Coordinates

To do so, we use the following relations,

$$r\dot{r} = x\dot{x} + y\dot{y} \quad ; \quad r^2\dot{\theta} = x\dot{y} - y\dot{x} .$$

We begin with the computation for $\dot{r}$.

$$\begin{aligned} r\dot{r} &= x\dot{x} + y\dot{y} \\ &= x(x - y - x(x^2 + 5y^2)) + y(x + y - y(x^2 + y^2)) \\ &= x^2 - xy - x^2(x^2 + 5y^2) + xy + y^2 - y^2(x^2 + y^2) \\ &= x^2 + y^2 - x^2(x^2 + 5y^2) - y^2(x^2 + y^2) \end{aligned}$$

Then, using $x^2 + y^2 = r^2$ and $\theta = \arctan(y/x)$, we find that

$$\begin{aligned} \implies r\dot{r} &= r^2 - r^2 \cos^2 \theta (r^2 \cos^2 \theta + 5r^2 \sin^2 \theta) - r^2 \sin^2 \theta (r^2) \\ &= r^2 - r^4 \sin^2 \theta - r^4 \cos^4 \theta - 5r^4 \sin^2 \theta \cos^2 \theta \\ &= r^2 - r^4 \sin^2 \theta - r^4 (1 - \sin^2 \theta) \cos^2 \theta - 5r^4 \sin^2 \theta \cos^2 \theta \\ &= r^2 - r^4 \sin^2 \theta - r^4 \cos^2 \theta + r^4 \sin^2 \theta \cos^2 \theta - 5r^4 \sin^2 \theta \cos^2 \theta \\ &= r^2 - r^4 (\sin^2 \theta + \cos^2 \theta) - 4r^4 \sin^2 \theta \cos^2 \theta \\ &= r^2 - r^4 - 4r^4 \sin^2 \theta \cos^2 \theta \\ \implies \dot{r} &= r - r^3 - 4r^3 \sin^2 \theta \cos^2 \theta \end{aligned}$$

Hence, this simplifies greatly into an expression we can use for the application of Poincaré–Bendixson Theorem. A similar calculation for $\dot{\theta}$, using the above identity, yields (in terms of Cartesian coordinates):

$$r^2\dot{\theta} = x^2 + y^2 - xy(x^2 + y^2) + xy(x^2 + 5y^2) .$$

Substitutions yield

$$r^2(1 - r^2 \cos \theta \sin \theta) + r^2 \cos \theta \sin \theta (r^2 \cos^2 \theta + 5r^2 \sin^2 \theta) ,$$

which reduces to

$$r^2\dot{\theta} = 4r^2 \left(r^2 \sin^3 \theta \cos \theta + \frac{1}{4} \right) .$$

Then, we have that

$$\begin{aligned} \dot{r} &= r - r^3 - 4r^3 \sin^2 \theta \cos^2 \theta \\ \dot{\theta} &= 4r^2 \sin^3 \theta \cos \theta + 1 \end{aligned}$$

3.3 (c) Determine the circle of maximum radius r_1 centered on the origin such that all trajectories have a radially *outward* component on the circle.

To do so, we seek the maximum value of r for which

$$\dot{r} = r - r^3 - 4r^3 \sin^2 \theta \cos^2 \theta > 0 \quad \forall \theta \in [0, 2\pi] .$$

It is important to note that the function of θ satisfies

$$0 \leq \sin^2 \theta \cos^2 \theta \leq \frac{1}{4} .$$

Therefore, by assuming this expression to be equal to the (worst-case) value of $1/4$, the above condition simplifies to

$$r - r^3 - r^3 > 0 \quad \implies \quad r > 2r^3 \quad .$$

The maximum r to satisfy this is then

$$r_{\max} = \frac{1}{\sqrt{2}} \quad .$$

In other words, any circle of radius less than $r_{\max}$ satisfies the property that the vector field in the r coordinate is always positively (or outward) oriented. In a strict sense, $r_{\max} = \frac{1}{\sqrt{2}} - \epsilon$ for some tiny positive number $\epsilon \ll 1$.

3.4 (d) Determine the circle of minimum radius r_2 centered on the origin such that all trajectories have a radially *inward* component on the circle.

We follow analogously to the above calculation, but now seeking the *minimum* r such that

$$\dot{r} = r - r^3 - 4r^3 \sin^2 \theta \cos^2 \theta < 0 \quad \forall \theta \in [0, 2\pi] \quad .$$

We now seek the worst-case value of the function in θ that *minimizes* the term, that is, we let the term including functions of θ vanish. Then, we have

$$r - r^3 < 0 \quad \implies \quad r < r^3 \quad .$$

This is satisfied $\forall r > 1$, so the minimum r (or the boundary of the closed set of r values that satisfy this condition, taking minimum in the limiting sense) is then simply

$$r_{\min} = 1 \quad .$$

In a strict sense, $r_{\min} = 1 + \epsilon$ for some tiny positive number $\epsilon \ll 1$.

3.5 (e) Prove that the system has a limit cycle somewhere in the trapping region $r_1 \leq r \leq r_2$.

Our proof will rely exclusively on the Poincaré–Bendixson Theorem. This theorem guarantees a limit cycle in some closed, bounded subset of the phase plane R . We have constructed this set as an annular region in the polar phase plane, with $R := (r, \theta) \in [r_{\max}, r_{\min}] \times [0, 2\pi]$. This is clearly a closed and bounded subset, satisfying the first criterion of the theorem. Secondly, our vector field is continuously differentiable on an open set containing R . Since in global Cartesian coordinates, the latter part regarding the domain is automatically satisfied, we use the fact that our expressions for $\dot{x}$ and $\dot{y}$ are polynomials, which immediately asserts that they are continuously differentiable. The third condition of the theorem states that R must not contain any fixed points. The only fixed point of the system is at the origin, and since $r_{\max} > 0$, this fixed point is excluded from our set R .

The final condition that must be satisfied for the theorem to apply is that there exists a trajectory which is confined in R for all time. In the above (sections 3.3 and 3.4), we have shown that the vector field points “outward” *everywhere* along the inner boundary of the annular region R , and points “inward” *everywhere* along the outer boundary of the region. This implies that trajectories within the region R that “creep” towards the edges of the annulus are directed back away from the boundary of R , and remain within the set. Therefore, we have already essentially shown that this final condition is satisfied.

Ergo, the annular region R contains a *closed orbit*, and all other trajectories within R tend towards this closed orbit. Thus, a limit cycle exists within the trapping region R .

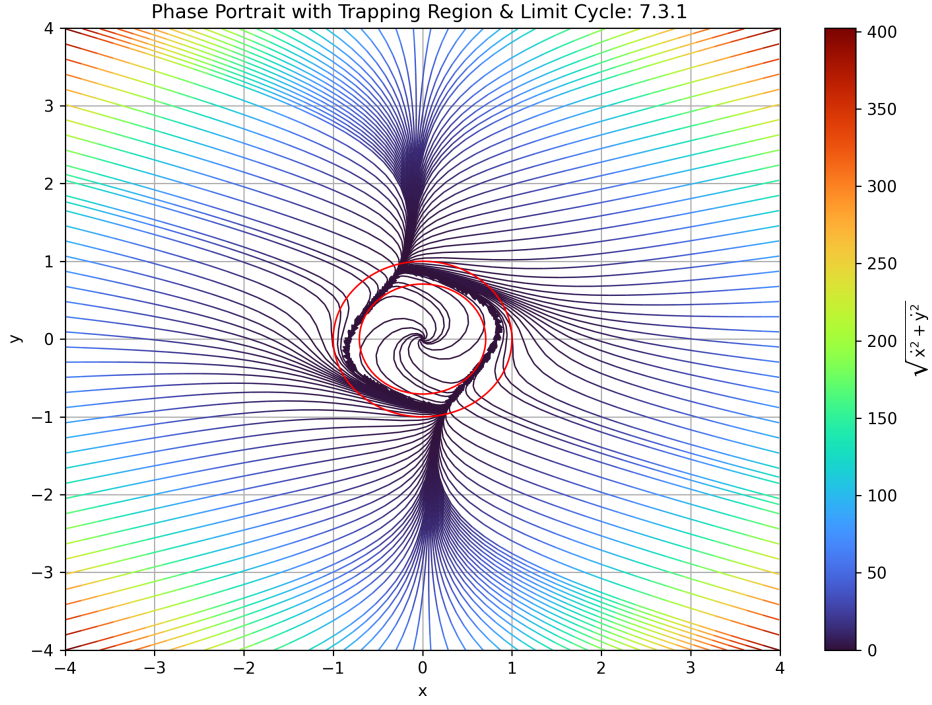


Figure 6: Phase portrait for the system in section 3.1. The trapping region bounding the limit cycle into a radial annulus is shown in red.

Note that if you wish to have a *strict* trapping region (i.e., not only the inner boundary of the trapping region), then the radial values should be modified:

$$R := (r, \theta) \in [r_{\max} - \epsilon, r_{\min} + \epsilon] \times [0, 2\pi] \quad \epsilon = e^{-10000}$$

This is all verified graphically by plotting the phase portrait for the system and coloring the boundaries of the trapping region. This is done in Figure 6.

4 Software Usage

The entirety of the work was carried out on pen and paper prior to typesetting, except for the plotting: this was completed using python (`numpy` and `matplotlib`).